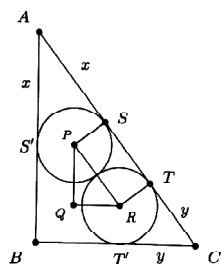


AMTI (NMTC) - 2014
RAMANUJAN CONTEST - INTER LEVEL

PART - A

1. The number of non-zero integers x such that $-10 < x < 15$ for which $3x^3 + 7x^2$ is the square of an integer is
(A) 5 (B) 6 (C) 0 (D) 1
2. The vertices of a triangle ABC are lattice points (that is, points with integer coordinates). Two of its sides have lengths which belong to the set $\{\sqrt{19}, \sqrt{2013}, \sqrt{2014}\}$. The maximum possible area of the triangle is
(A) 2013 (B) 1007 (C) 2014 (D) $\sqrt{2014}$
3. The minimum value of the terms of the sequence $\sqrt{\frac{7}{6}} + \sqrt{\frac{96}{7}}, \sqrt{\frac{8}{6}} + \sqrt{\frac{96}{8}}, \sqrt{\frac{9}{6}} + \sqrt{\frac{96}{9}}, \dots, \sqrt{\frac{95}{6}} + \sqrt{\frac{96}{95}}$ is
(A) 6 (B) 7 (C) 8 (D) 4
4. ABCD is a cyclic quadrilateral in a circle of radius r . AB is a diameter of the circle. CD is parallel to AB. $CD = b$, $AD = BC = a$. The value of $\frac{2r^2 - a^2}{br}$ is
(A) 1 (B) 2 (C) 3 (D) 4
5. Positive integers a, b, c are chosen so that $a < b < c$ and the system of equations $2x + y = 2013$ and $y = |x - a| + |x - b| + |x - c|$ have exactly one solution. Then the minimum value of c is
(A) 760 (B) 1007 (C) 2013 (D) 2012
6. ABCD is a rectangle. P and Q are points on AB and BC respectively such that the area of triangle APD = 5, area of triangle PBQ = 4 and area of triangle QCD = 3, all areas in square units. Then the area of the triangle DPQ is square units is
(A) 12 (B) $\frac{20}{3}$ (C) $2\sqrt{21}$ (D) $\sqrt{21}$
7. The number of right triangles of integer length sides and the product of the leg lengths is equal to three times the perimeter is
(A) 0 (B) 1 (C) 2 (D) 3
8. Let S be the set of ordered triples (x, y, z) of real numbers for which $\log_{10}(x + y) = z$ and $\log_{10}(x^2 + y^2) = z + 1$. a, b are real numbers such that for all ordered triples (x, y, z) in S , we have $x^3 + y^3 = a \cdot 10^{3z} + b \cdot 10^{2z}$. Then the value of $(a + b)$ is
(A) $\frac{15}{2}$ (B) $\frac{29}{2}$ (C) 15 (D) $\frac{39}{2}$
9. The internal bisector of $\angle A$ of a triangle ABC meets BC at P and $b = 2c$. If $9AP^2 + 2a^2 = k \cdot c^2$, then k is
(A) 8 (B) 3 (C) 19 (D) 18
10. The product of 5 odd primes is a five digit number of the form $ab0ab$ where a, b are digits and the hundreds digit is zero. The number of such numbers is
(A) 0 (B) 9 (C) 13 (D) 18
11. The number of perfect square divisors of the product $1! 2! 3! 4! \dots 9!$ is
(A) 583 (B) 615 (C) 627 (D) 672
12. Srilekha, Priyanka, Vidya and Vishwa bought a gift for their classmate's birthday. The gift is with someone of the four. Each makes a statement. One statement is false and the other three are true.
Srilekha : I do not have the gift and Vidya does not have the gift.
Priyanka : I do not have the gift and Srilekha does not have the gift.
Vidya : I do not have the gift and Priyanka does not have the gift.
Vishwa : I do not have the gift and Srilekha does not have the gift.
The gift is with
(A) Srilekha (B) Priyanka (C) Vidya (D) Vishwa

13. In the figure below the triangle is a 3 – 4 – 5 sides triangle. Two equal circles are placed as in the figure. The radius of each circle is

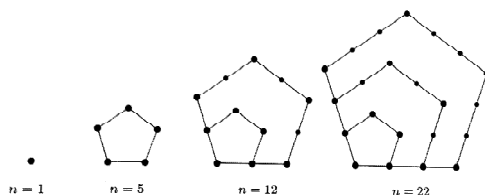


- (A) $\frac{4}{5}$ (B) $\frac{5}{8}$ (C) $\frac{6}{11}$ (D) $\frac{5}{7}$

14. The number of positive integer pairs (m, n) where n is odd, that satisfy $\frac{1}{m} + \frac{4}{n} = \frac{1}{12}$ is
 (A) 1 (B) 2 (C) 3 (D) 11
15. The number of points common to the two graphs whose equations are $||x| - |y|| + |x| + |y| = 2$ and $2y = |2x - 1| - 3$ is
 (A) 1 (B) 2 (C) 3 (D) 4

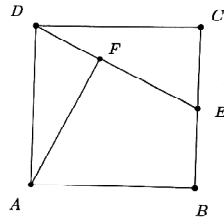
PART - B

16. When simplified, the value of $\frac{(10^4 + 324)(22^4 + 324)(34^4 + 324)(46^4 + 324)(58^4 + 324)}{(4^4 + 324)(16^4 + 324)(28^4 + 324)(40^4 + 324)(52^4 + 324)}$ is _____.
17. The least positive integer a such that the equation $\cos^2 \pi (a - x) - 2 \cos \pi (a - x) + \cos \frac{3\pi x}{2a} \cdot \cos \left(\frac{\pi x}{2a} + \frac{\pi}{3} \right) + 2 = 0$ has a real root is _____.
18. The quadrilateral ABCD is inscribed in a circle. The diagonals AC and BD intersect at Q. The sides DA and CB are produced to meet at P. Given the $CD = CP = DQ$, the value of $\angle CAD =$ _____.
19. Dividing a three digit number by the number obtained from it by swapping its first and last digits we get 3 as the quotient and the sum of the digits of the original number as the remainder. The number of such 3-digit numbers is _____.
20. The number of pairs (a, b) of real numbers such that the zeros of the polynomials $(6x^2 - 24x - 4a)$ and $(x^3 + ax^2 + bx - 8)$ are all non-negative real numbers is _____.
21. The number of pairs of positive integers (m, n) such that $m \times n = \gcd(m, n) + \text{lcm}(m, n)$ is _____.
22. Two spheres of radii 3 cm and 4 cm, whose centres are 5 cm apart cut each other. The volume common to both the spheres is given by $\frac{a}{b} \pi \text{ cm}^3$ where $\frac{a}{b}$ is the fraction with the lowest numerator and denominator, then the value of $\frac{a}{b}$ is _____.
23. The number of points with integer coordinates on the line $x + y = 1$ which are located inside the circle with centre (0, 0) and radius 3 is _____.
24. Look at the pattern shown in the figure Pentagonal Numbers.



25. The number of natural numbers $n \leq 2013$ for which $2^n - 1$ is divisible by 7 is _____.
26. A fruit basket contains 4 bananas, 5 oranges and 6 apples. The number of ways a person can make a selection of the fruits from among the fruits in the basket is _____.

27. a, b, c are positive integers such that $a + b + c = 2013$. Given that $a! b! c! = m10^n$ where m, n are integers and m is not divisible by 10, the smallest value of n is _____.
28. In the figure (i) below, $ABCD$ is a square. E is the midpoint of CB . AF is drawn perpendicular to DE . If the side of the square is 2013 cm the length of FB is _____ cm.



29. The number of primes p for which $(p + 2)$ and $p^2 + 2p - 8$ are both primes is _____.
30. $f(x)$ is a linear function. $f(0) = -5$ and $f(f(0)) = -15$. The number of values of α for which the solutions to the inequality $f(x) f(\alpha - x) > 0$ form an interval of length 2 is _____.

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